MIL-B-48570A (AR)
30 OCTOBER 1979
SUPERSEDING
MIL-B-48570 (MU)
12 May 1975

MILITARY SPECIFICATION

BAND, CENTERING FOR PROJECTILE 105MM DISCARDING SABOT

This specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.-This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of a centering band to be used in the projectile, 105mm discarding sabot.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents.-The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-392 - Plastic Molding Material, Acetal, Injection and Extrusion

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

FSC: 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunitions Components

DRAWINGS

11820393 (SH.1) - Band, Centering (Machined)
11820393 (SH.2) - Band, Centering (Blank)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-Materials and parts shall comply with the requirements specified in the applicable drawings and referenced specifications.

3.2 Raw material

3.2.1 Mechanical properties.-The mechanical properties shall comply with the requirements of the applicable drawing, the applicable specification, and federal specification L-P-392 (Plastic Molding Material, Acetal, Injection and Extrusion) with the exception of dielectric constant and para. 4.4.1.3 therein.

3.3 Band blank

3.3.1 Roll deflection.-Band blanks, prior to stress relief anneal, shall not fracture, crack or break while under load.

3.3.2 Tensile strength.-Band blanks, after stress relief anneal, shall have a minimum breaking load of 2450 pounds (1111kg).

3.3.3 Elongation.-Band blanks elongation, after stress relief anneal, shall be a minimum of .091 inches (2.31mm).

3.3.4 Moisture content.-Band blanks moisture content shall be 0.5% max.

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3.4 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship - machined band.-The requirements for workmanship are shown on the applicable drawing, on the applicable specification and the following:

3.5.1 Material defects.-Soundness of material shall be as required by the applicable specification and drawing. All components shall be free from blisters, cracks, check, foreign inclusions, scratches, voids and any surface or internal defect which may detrimentally affect serviceability or appearance in the judgement of the appropriate technical agency.

3.5.2 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawings and shall be in accordance with the provisions of MIL-A-2550.

3.5.3 Burr.-No component shall have a burr which might interfere with the assembly or functioning of the projectile or which might be injurious to personnel handling the item.

3.5.4 Flash.-Machined bands must be free of flashes.

3.5.5 Foreign matter.-All parts shall be free of chips, dirt, grease, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

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4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the items in sample quantities as indicated below.

Part Description	Drawing	Quantity
Raw Material (Plastic-Delrin 500)	11820393	25 lbs.
Band, Blanks (Prior to Stress Relief)	11820393	25
Band, Blanks (After Stress Relief)	11820393	50
Band, Machined	11820393	25

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
	Band, Centering		1 OF 1	11820393 (Sh, 1 & 2)
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
I	Raw Material				
	A. Mechanical properties	(25 pounds)			
	B. testing per L-P-392	(11.34kg)		3.2.1	4.5.2
II	Band, Blanks				
	A. Roll deflection test (prior to stress relief)	25		3.3.1	4.5.3
	B. Examination for defects & Moisture content	25		3.1 & 3.3.4	4.4.2.1 & 4.5.5
	C. Breaking load and elongation test (after stress relief)	25		3.3.2& 3.3.3	4.5.4
III	Band, Machined	25			
	A. Examination for defects			3.1	4.4.2.2
	B. Material defects			3.5.1	4.5.6

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.1.1 Band, blanks, lot.-A band blank lot shall consist of blanks which have been processed under the same conditions. A band blank lot shall contain only one material batch.

4.4.1.1.1 Batch.-A batch of raw material shall be defined as a unit of product prepared for shipment, and may consist of a uniform blend of two or more "production runs" of acetal of the same type, class and color.

4.4.1.2 Band, machined, lot.-A machined band lot shall be made from only one band blank lot.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plan.-Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-48570A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Band, Centering, Blank		11820393 (SH. 2)
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Roll deflection test (prior to stress relief)		100%	3.3.1	4.5.3
102	Breaking load and elongation test	(See Note)		3.3.2 and 3.3.3	4.5.4
103	Moisture content	(See Note)		3.3.4	4.5.5
Minor					
201	Diameter, outside		0.65%	3.1	Gage
202	Diameter, inside		0.65%	3.1	Gage
203	Length, total		0.65%	3.1	Gage

NOTE: Ten band blanks from each lot after stress relief anneal shall be tested for compliance with the requirements. Failure of two or more samples shall be cause for rejection of the lot. If one sample fails, ten additional blanks shall be tested. Failure of one or more of these additional samples will cause the lot to be rejected.

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-48570A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.2	Band, Centering, Machined				11820393 (SH. 1)
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of counterbore with pitch diameter of thread		0.40%	3.1	Gage
102	Pitch diameter of thread		0.40%	3.1	Gage
103	Minor diameter of thread, max.		0.40%	3.1	Gage
104	Diameter of counterbore		0.40%	3.1	Gage
105	Diameter, outside, min.		0.40%	3.1	Gage
106	Diameter of chamfer, rear face		0.40%	3.1	Gage
107	Diameter of chamfer, front face		0.40%	3.1	Gage
108	Angle of chamfer at outside diameter, front face (if applicable)		0.40%	3.1	Gage
109	Angle of chamfer at outside diameter, rear face		0.40%	3.1	Gage
110	Length total		0.40%	3.1	Gage
111	Material defects		0.40%	3.1	4.5.6

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.2	Band, Centering, Machined (Cont'd)				11820393 (SH. 1)
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Depth of counterbore		0.65%	3.1	Gage
202	Angle at rear of thread cavity, incorrect		0.65%	3.1	Visual
203	Chamfer at rear of counterbore, missing or incorrect		0.65%	3.1	Visual
204	Surface finish improper		0,65%	3.1	Visual
205	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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4.4.3 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Testing conditions.-Unless otherwise specified in the test method or applicable specification, the atmospheric conditions surrounding the specimen prior to and during tests shall be 23°C ± 2°C and 50 ± 4 percent relative humidity.

4.5.2 Mechanical properties - Raw material.-Raw material will be tested for compliance with 3.2.1.

4.5.3 Roll deflection test.-Each band blank will be placed in a fixture capable of rotating the band on its circumference through 180° (π rad.) between a curved track and a wheel rotating at 180 rpm (18.85 rad./sec) (see figure 1). While being rotated the band will be compressed by 15% based on the nominal internal diameter of the band blank. Band blanks will be visually inspected 100% and any evidence of a fracture, crack or break will be cause for rejection.

4.5.4 Breaking load and elongation test.-Each band blank submitted to this test will be placed in a saddle type specimen holder of the half moon internal design. The radius of curvature of each half moon must be equal to the nominal internal radius of the band blanks. The saddle shall be pulled apart until in contact with the specimen. This point will be the reference zero point. The distance between the crossheads shall be measured and recorded as initial gage length. The saddle shall then be pulled apart at the rate of two inches per minute (.85 mm/sec). The load at break shall then be recorded. The distance between the crossheads shall be measured again to determine final gage length.

ELONGATION = Final gage length - Initial gage length.

4.5.5 Moisture content test.-Each band blank submitted to this test shall be weighed to the nearest milligram. Then the band blanks will be heated in an air circulating oven at 105°C ± 2°C for twenty-four hours, and cooled in a desicator over anhydrous calcium chloride. The band blanks

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will then be reweighed to the nearest milligram. After weighing, the band blanks will be heated in an air circulating oven at 105°C ± 2°C for one hour, and cooled in a desiccator over anyhdrous calcium chloride and weighed to the nearest milligram. The one hour heating and weighing cycle will be repeated until two consecutive weights are within one milligram. Percent moisture content shall be calculated as follows:

$$\% \text{ MOISTURE} = \frac{\text{Initial weight} - \text{Final weight}}{\text{Initial weight}}$$

4.5.6 Material defect - Machined bands.-Machined bands shall be inspected to determine compliance with 3.5.1. Each machined band selected for this inspection shall be tested by placing a light equivalent to GE 1493, clear, single filament, 6.5V, 2.75 amps, internally in the band. The machined band will be rotated 360° (2 rad) and inspected at (1) x magnification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking

5.1.1 Level A - Military package.-NA

5.1.2 Level B - Minimum military package.-Preservation, packaging, packing and marking shall comply with the applicable drawing, specification, military specification MIL-STD-1169 and this specification.

5.2 Packing.-Centering band components shall be packed in a manner which will insure acceptance by common carrier and will afford protection against physical damage during shipment from the supply source to the first receiving activity for immediate use. Shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.3 Marking.-In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Shipping.-When components from more than one lot are shipped as a car load, each lot must be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

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6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Ordering data.-See MIL-A-48078.

a. Data Cards - Data cards shall be prepared for each lot in accordance with MIL-STD-1167.

b. Delivery of data cards to the technical agency.-The following address is to be added to those listed in the contract (Form DD 1423) for the delivery of one copy, to the technical agency of the Ammunition Data Cards:

Commander, US Army Armament
Research and Development Command
ATTN: DRDAR-QAR-E
Dover, New Jersey 07801

6.3 Submission of inspection equipment designs for approval.-
See MIL-A-48078. Submit equipment designs, as required to:

Commander
US Army Research and Development Command
ATTN: DRDAR-QAR-I
Dover, NJ 07801

Custodian:
ARMY-AR

Preparing Activity:
ARMY-AR

Project No: 1315-A215

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